

Gabriel Cassoli Capellini

Address Institute of Astronomy, Geophysics and Atmospheric Sciences - University of São Paulo (IAG-USP)
Email R. do Matão, 1226 - Butantã, São Paulo - SP, 05508-090
Nationality gabriel.capellini@usp.br
Brazilian

Education

2025-present Master of Science, Department of Astronomy - University of São Paulo (IAG-USP), Brazil

Advisor Prof. Dr. Maria Victoria del Valle

Thesis:

Multiwavelength Modeling of Gamma-ray Emitters: Unveiling the Nature of Unidentified Sources

2020-2024 Bachelor of Science, Astronomy - University of São Paulo (IAG-USP), Brazil

Advisor Prof. Dr. Maria Victoria del Valle

Thesis:

T Tauri stars as Transient Gamma-ray Sources

Published works

- 2026** Capellini, G.C. and del Valle, M.V. : *Non-thermal leptonic emission from a protoplanetary nebula*
accepted for publication in Proceedings of Science HEPRO IX.
- 2026** Capellini, G.C. and del Valle, M.V. : *Are T Tauri stars gamma-ray emitters? Insights from a non-thermal model*
accepted for publication in Boletim da SAB volume 37.
- 2026** del Valle, M.V. and Capellini, G.C. : *Hadronic Transient Gamma-ray Emission from a Protostar*
submitted to Astronomy & Astrophysics.
- 2025** Capellini, G.C. and del Valle, M.V. : *T Tauri stars as transient gamma-ray sources.*
Boletim da SAB volume 36.
- 2024** del Valle, M.V. and Capellini, G.C. : *On the nature of the transient gamma-ray source associated with a protostar*
accepted for publication in Journal of the Italian Astronomical Society, MemSAIt
- 2024** Capellini, G.C. and del Valle, M.V. : *Estrelas T Tauri como fontes transientes de raios gama.*
Instituto de Astronomia, Geofísica e Ciências Atmosféricas - Universidade de São Paulo (IAG-USP)

Conferences and Workshops

- Oct.2025** XLVIII Annual Reunion of the Brazilian Astronomical Society -Caxambu, Brazil - RASAB XLVIII
Poster Are T Tauri stars gamma-ray emitters? Insights from a non-thermal model
- Oct.2025** Astronomia ao meio-dia -São Paulo, Brazil
Talk Galactic Accelerators: the Search for Unidentified Sources

Sep.2025	X IAG Science Day -São Paulo, Brazil - IAG Science Day Talk Jets in Planetary Nebulae: a New Window to Gamma-ray Sources
Aug.2025	High Energy Phenomena in Relativistic Outflows IX -Rio de Janeiro, Brazil - Hepro-IX Poster A new gamma-ray source? The role of jet-outflow interaction in a protoplanetary nebula
Aug.2025	CLAF/ICTP-SAIFR Latin-American Astroparticle Physics School -São Paulo, Brazil Poster The role of a jet-outflow system in gamma-ray emission from a protoplanetary nebula
Oct.2024	32° USP International Symposium on Scientific and Technological Initiation -São Paulo, Brazil - 32° SIICUSP Poster T Tauri stars as transient gamma-ray sources First Stage and International Stage
Sep.2024	XLVII Annual Reunion of the Brazilian Astronomical Society -Águas de Lindoia, Brazil - RASAB XLVII Poster T Tauri stars as transient gamma-ray sources
Sep.2024	IX IAG Science Day -São Paulo, Brazil - IAG Science Day Talk T Tauri Stars as Transient Gamma-ray Sources
Jun.2024	The first MPIK-CDY School on the Future of Gamma-Ray Astronomy -Heidelberg, Germany - MPIK-CDY Poster Particle Acceleration in Galactic Sources
Nov.2023	VIII IAG Science Day -São Paulo, Brazil - IAG Science Day Talk Particle Acceleration in Galactic Sources
Oct.2023	31° USP International Symposium on Scientific and Technological Initiation -São Paulo, Brazil - 31° SIICUSP Poster & Talk Particle Acceleration in Galactic Sources First Stage and International Stage
Oct.2023	XLVI Annual Reunion of the Brazilian Astronomical Society- Rio de Janeiro, Brazil- RASAB XLVI Poster Study on Particle Acceleration in Shock Waves in Galactic Sources

Research experience

Mar.2025 - present	University of São Paulo (IAG-USP), São Paulo, Brazil <i>Multiwavelength Modeling of Gamma-ray Emitters: Unveiling the Nature of Unidentified Sources</i> Supported by CAPES (88887.142048/2025-00) and FAPESP (No.2025/06590-1)
Feb.2024 - Dec.2024	University of São Paulo (IAG-USP), São Paulo, Brazil <i>T Tauri Stars as Transient Gamma-ray Sources</i> Supported by FAPESP (No.2022/16530-8)
Feb.2023 - Dec.2023	University of São Paulo (IAG-USP), São Paulo, Brazil <i>Particle Acceleration in Galactic Sources</i> Supported by FAPESP (No.2022/16530-8)

Teaching experience

Feb.2022 - Jul.2022	University of São Paulo (IAG-USP), Brazil <i>Teaching Assistant in Introduction to Astronomy I - USP</i>
Feb.2024 - Jul.2024	University of São Paulo (IAG-USP), Brazil <i>Teaching Assistant in Introduction to Astronomy I - USP</i>
Aug.2024 - Dec.2024	University of São Paulo (IAG-USP), Brazil <i>Teaching Assistant in High Energy Astrophysics - USP</i>